LESSON 01 · CHEAT SHEET

Numbers, Text & Formatting

VARIABLES & TYPES

```
salary = 32500          # int    whole number
hours = 37.5            # float  decimal
name = "Jane Smith"     # str    text
is_member = True        # bool   True / False
manager = None          # "no value" (VBA: Nothing)
type(hours)             <class 'float'>
```

ARITHMETIC

`17 / 7`	2.428...	always a float
`17 // 7`	2	whole part (INT)
`17 % 7`	3	remainder (MOD)
`2 ** 3`	8	power (Excel: 2^3)
`round(2.718, 2)`	2.72	ROUND
`0.1 + 0.2`	0.30000000000000004	normal, round at the end

TEXT (STRINGS)

`"Jane" + " " + "Smith"`	join (VBA: `&`)
`" hi ".strip()`	"hi" (TRIM)
`"jANE".title()`	"Jane" (PROPER)
`"Hi".upper()` / `.lower()`	UPPER / LOWER
`"a-b".replace("-", "")`	"ab" (SUBSTITUTE)
`len("Jane")`	4 (LEN)
`"FIN-42".find("-")`	3 (FIND minus 1)
`"-" * 5`	"-----" (REPT)

Chain them: `" emp-1 ".strip().upper()` → `"EMP-1"`

SLICING: LEFT, RIGHT, MID

```
text = "FIN-0042"
        01234567    ← positions start at 0
text[0]       "F"       # first character
text[-1]      "2"       # last character
text[:3]      "FIN"     # LEFT(text, 3)
text[-4:]     "0042"    # RIGHT(text, 4)
text[4:6]     "00"      # MID(text, 5, 2)
```

`text[a:b]` runs from a up to, **not including**, b. It has b−a characters.

CONVERTING

```
int("42")           42
float("37.5")       37.5
str(42)             "42"
float("£1,500".replace("£", "").replace(",", ""))
                    1500.0
" Yes ".strip().lower() == "yes"     True
```

F-STRINGS: NUMBER FORMATS

```
name = "Jane"; pay = 2708.333
f"Hi {name}!"          Hi Jane!
```

`{pay:.2f}`	2708.33	2 decimals
`{pay:,.2f}`	2,708.33	+ thousands
`{32500:,}`	32,500	thousands
`{0.125:.1%}`	12.5%	percent
`{42:04d}`	0042	zero-padded
`{"ab":<6}`	"ab "	left in 6
`{"ab":>6}`	" ab"	right in 6

TRUE OR FALSE

```
salary == 28000       # equal (VBA: =)
salary != 28000       # not equal (VBA: <>)
25000 <= salary <= 32000   # between, limits included
hours >= 35 and rating >= 3
not is_member
```

VBA → PYTHON

`Dim x As Long`	`x = 5`
`' comment`	`# comment`
`Debug.Print x`	`print(x)`
`CInt / CDbl / CStr`	`int() / float() / str()`
`Format(x, "#,##0.00")`	`f"{x:,.2f}"`
`And / Or / Not`	`and / or / not`

GOTCHAS

- `^` is not power. Use `**`
- `=` stores, `==` compares
- `Salary` ≠ `salary` (case-sensitive)
- `"5" + 3` is an error. Convert first
- `round(2.5)` is 2: halves round to even, like VBA's `Round`